MSc Project Report 2025-26

Examining the Timeliness of Routine Vaccination During Heatwaves.

Candidate Number: 493736

Page count: 0

Project length: Standard

Submitted in part fulfilment of the requirements for the degree of MSc in Medical Statistics

Date of submission: July 2026

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Abstract

All project reports should include a structured Abstract, not exceeding 300 words, on a single standalone page just after the Contents. This should appear before the main body of the project report (which will start with the introduction).

The Abstract may be structured into four key sections: - Background – summarising the problem being considered. - Methods – describing how the study was performed. - Results – listing the salient results. - Conclusions – stating the principal conclusions.

Acknowledgements

Your project report should include an Acknowledgements section, before the Introduction to your work proper. Detailed guidance on what to put in the Acknowledgements section is given in Section 12.1

1 Introduction

1.1 Background

Globally, extreme weather events such as floods, cold waves, droughts and heatwaves have become increasingly frequent and intense(1). These events disrupt multiple sectors including energy infrastructure, agriculture, food production, healthcare and livelihoods, resulting in substantial social and economic losses (2,3). Consistent with these changes, global mean near-surface temperatures have risen by approximately 1.4–1.5°C above pre-industrial (1850–1900) levels, with the five-year average temperature during 2020–2025 being the highest on record and 2024 recorded as the warmest year globally (4). This continued warming is largely driven by increasing greenhouse gas emissions, together with changes and interactions in atmospheric and oceanic circulation (4,5). One of the most direct consequences of rising global temperatures is the increasing occurrence of heatwaves, commonly defined as periods of unusually high temperatures that persist for several consecutive days. Climate projections indicate that both the frequency and intensity of heatwaves will continue to increase unless substantial reductions in greenhouse gas emissions are accompanied by investments in climate adaptation and resilient infrastructure (5).

Heatwaves have important implications for population health and healthcare systems. High temperatures have been associated with challenges in maintaining the cold chain required for vaccines, medicines and medical equipment (6,7), increased incidence of heat-related illnesses and injuries, particularly among older adults (8), prolonged recovery among hospitalized patients, increased in-patient mortality (9,10), and excess mortality across many regions of the world (11–13). Beyond their direct effects on health, these impacts place additional strain on already constrained health systems.

Nigeria has experienced several notable heatwave events in recent decades, including those occurring in 2002, 2019, 2020, 2023 and 2024 (14–16). These events have been linked to broader climatic changes, compounded by rapid urbanisation, deforestation and land-use change. Their impacts have been felt across multiple sectors. In agriculture, the 2024 heatwave was associated with an estimated 20% decline in agricultural production in a country where food production depends heavily on rainfall (16,17). Heatwaves have also increased pressure on Nigeria's energy infrastructure through higher cooling demand, contributing to electricity shortages and national grid failures(17). In the health sector, they have been associated with increased occurrence of heat-related illnesses such as heat stroke and meningitis, while environmentally they have contributed to increased wildfire risk and the progression of desertification, particularly in northern Nigeria (17,18).

Nigeria's National Programme on Immunization (NPI) was established in 1978 to provide routine immunization services for children. Over the years, the programme has expanded to include additional vaccine antigens in line with the country's Expanded Programme on Immunization (EPI). According to the current national immunization schedule, a child is considered fully immunized upon receiving one dose of Bacillus Calmette–Guérin (BCG), the hepatitis B birth dose, four doses of oral polio vaccine (OPV), two doses of inactivated polio vaccine (IPV), three doses each of the pentavalent, pneumococcal conjugate (PCV) and rotavirus vaccines, two doses of measles-containing vaccine (MCV), and one dose each of the yellow fever and meningitis vaccines (19). Rotavirus vaccine is the most recent addition to the national schedule, having been introduced in phases during 2022 (19). The current childhood immunization schedule is presented in Table 1.1.

Despite substantial progress over recent decades, routine childhood immunization coverage in Nigeria remains below national and global targets and is characterised by marked geographical and socioeconomic disparities. Nationally, the proportion of children receiving all basic vaccinations increased from 13% in 2003 to 39% in 2024 (19). However, vaccination coverage varies considerably across the country, with consistently lower coverage observed in northern Nigeria compared with the southern regions (20,21). The 2024 NDHS similarly reported the highest coverage in the South South region (36%) and the lowest in the North West region (12%) (19). These disparities have been associated with several factors, including place of delivery, maternal education, household wealth, and urban–rural residence (19,21,22).

Nigeria continues to account for a substantial proportion of the global burden of undervaccinated children(23). Persistent challenges affecting the immunization programme include supply chain constraints, inadequate healthcare personnel and funding, limited geographical access to health facilities, and vaccine hesitancy associated with sociocultural and religious factors (23,24). More recently, the COVID-19 pandemic further disrupted routine immunization services, slowing progress in vaccination coverage and access (25).

Table 1.1: Nigeria's National Programme on Immunization schedule.

Minimum target age	Vaccine
At birth	BCG
	OPV 0 (Birth dose)
	Hepatitis B birth dose
6 weeks	Pentavalent (DPT, Hepatitis B, HiB) 1
	Pneumococcal Conjugate Vaccine 1
	OPV 1 (1st dose)
	Rotavirus 1
	IPV1
10 weeks	Pentavalent (DPT, Hepatitis B, HiB) 2
	Pneumococcal Conjugate Vaccine 2
	OPV 2 (2nd dose)
	Rotavirus 2
14 weeks	Pentavalent (DPT, Hepatitis B, HiB) 3
	Pneumococcal Conjugate Vaccine 3
	OPV 3 (3rd dose)
	Rotavirus 3
	IPV2
6 months	Vitamin A (1st dose)
9 months	Measles 1st dose (MCV1)
	Yellow Fever
	Meningitis Vaccine
12 months	Vitamin A (2nd dose)
15 months	Measles 2nd dose (MCV2)

Although considerable attention has been given to vaccination coverage in Nigeria, considerably less is known about whether children receive vaccines at the recommended ages(26,27). Vaccination timeliness is important because delays prolong children's susceptibility to vaccine-preventable diseases despite eventual vaccine receipt.

1.2 Aims and Objectives

2 Literature review

2.1 Vaccine timeliness

Several country-specific studies, multi-country analyses, and systematic reviews have examined the timeliness of routine childhood vaccination using nationally representative survey data, primarily from the Demographic and Health Surveys (DHS) and Multiple Indicator Cluster Surveys (MICS) (27–35). The largest and most recent systematic review synthesised evidence from 224 studies conducted over four decades (1978-2024) across low- and middle-income countries (35). Despite this growing body of evidence, systematic reviews have highlighted important methodological limitations and substantial heterogeneity in the approaches used to assess vaccination timeliness (33,35).

Across all studies, vaccination timeliness has generally been measured using one of two approaches. The first categorizes vaccination into early, timely and delayed administration or delayed versus timely administration(32,34,35), while the second treats delay as a continuous time-to-event outcome, typically defined as the number of days between the recommended age of vaccination and the observed vaccination date (28,30,31). Although both approaches are widely used, no universally accepted definition of vaccine delay exists(35). Consequently, vaccination delay has been defined using a variety of thresholds based on the time elapsed since the recommended age for vaccination. Common thresholds include four days, one week, two weeks, four weeks (28 days), one month, two months, and ninety days after the scheduled vaccination age (33,35). Whether vaccine timeliness is analysed as a discrete or continuous outcome was found to have important implications for the choice of statistical models. Studies using categorized measures of timeliness employed logistic regression models, whereas those analyzing vaccination delay as a continuous time-to-event outcome typically used survival analysis methods, including the Cox proportional hazards model (33).

Another important source of methodological variation relates to the age range of children included in vaccine timeliness studies. A recent systematic review identified 35 different age ranges across 57 studies, although children aged 12–23 months were the most commonly studied population (33). Overall, the review found substantial inconsistency in age eligibility, with studies including children ranging from those younger than one year to those aged five years and above, while some studies did not clearly define the age range of participants. Such variation may influence reported estimates of vaccine timeliness, particularly for vaccines administered later during infancy such as the second measles containing vaccine, as younger children may not yet have reached the recommended age for all scheduled vaccinations, whereas older children have had a longer opportunity to receive delayed vaccines.

Studies have consistently identified several determinants of vaccination timeliness. The determinants considered across most studies are generally divided into child-level factors such as child's sex, birth order, place of birth, birth year, maternal and paternal factors such as age, level of education and occupation, household level factors such as religion, ethnicity, wealth index, residence type, travel times to facilities and health system factors such as health facility availability, accessibility and staffing(33,35). Across major studies, children living in rural areas, those from poorer households, and those whose mothers had little or no formal education were consistently more likely to experience delayed vaccination (27,34,35). These findings suggest that socioeconomic inequalities remain an important determinant of vaccine timeliness and highlight the need for interventions targeted towards disadvantaged populations (27). Further, vaccination was generally more timely for birth doses and earlier childhood vaccines, while delays increased for vaccines scheduled later in infancy (31,34). A recent analysis of more than 90 DHS and MICS surveys reported median timely vaccination coverage of approximately 79% for birth doses, declining to 70% at six weeks, 59% at ten weeks, 50% at fourteen weeks, 56% at nine months and 50% at fifteen months (31). This pattern suggests that children become progressively more likely to experience delayed vaccination as they progress through the routine immunization schedule. Beyond individual and household characteristics, several studies have attributed delayed vaccination to health system disruptions such as Covid-19 (36), health-system constraints, including vaccine stock-outs, inadequate supply chains, shortages of healthcare personnel, limited access to health facilities, and civil unrest or insecurity in some settings[xxx]. These factors may disrupt routine immunization services and reduce access to timely vaccination, particularly in resource-constrained settings.

Studies of vaccine timeliness have also differed in how they handled children with missing vaccination dates, which are required to calculate vaccination delay and timeliness. Some studies excluded these children from the analysis, whereas others used survival analysis techniques that accounted for incomplete information (35). For example, left censoring was used for children whose mothers reported that the vaccine had been received before the interview but without a recorded date, while right censoring was applied to children who had not yet received the vaccine by the time of the interview. However, few studies incorporated both left and right censoring.

These inconsistencies in age groups used, definitions and operationalization of vaccine delay, adjusted predictor variables and modelling approaches complicate comparisons and harmonization of vaccine timeliness estimates across studies and countries.

2.2 Heatwaves

Heatwaves are among the most widely studied climate extremes because of their increasing frequency, intensity and duration under climate change. Although there is no universally accepted definition of a heatwave, most definitions describe a heatwave as a period of abnormally high temperatures persisting for several consecutive days relative to the local climatology(37,38). Earlier studies often relied on absolute temperature thresholds (for example, daily maximum temperatures exceeding a fixed value), while more recent approaches favour percentile- or quantile-based definitions that account for local climatic conditions and population adaptation, such as temperatures exceeding the 90th, 95th or 99th percentile of the historical temperature distribution for a given location and time of year (37,39,40). Other studies have incorporated both temperature intensity and duration by defining heatwaves using moving averages of daily maximum temperatures or indices that combine daytime and nighttime temperatures. The evolution from fixed thresholds to location-specific and duration-sensitive definitions has substantially improved comparability across climatic regions and has enabled more robust assessments of heatwave occurrence under changing climatic conditions (37,40,41).

Alongside advances in heatwave definitions, considerable progress has been made in attributing individual heatwave events to human-induced climate change (37,39). Early attribution studies primarily compared observed temperatures with long-term climatological averages, whereas modern attribution frameworks combine observational records, climate model simulations and extreme value statistics to quantify changes in both the probability and intensity of extreme events. Event attribution is now commonly undertaken using probabilistic approaches such as the World Weather Attribution (WWA) framework, which compares the probability of an event occurring in the present-day climate with that in a counterfactual climate without human influence(39). These approaches frequently employ generalized extreme value (GEV) models fitted to annual temperature maxima to estimate return levels and return periods, allowing rare events to be characterised using exceedance probabilities or high quantiles of the fitted extreme value distribution. The resulting metrics, including probability ratios and fractions of attributable risk, provide quantitative evidence of the contribution of human-induced climate change to observed heatwave events. Recent methodological developments have highlighted the importance of accounting for non-stationarity, model uncertainty and physical consistency when estimating return levels and attributing extreme heat events under a warming climate(37,39,42). Recent work has further argued that traditional attribution methods require refinement to better account for compound climate processes, observational uncertainty and evolving background climatic conditions, particularly when projecting future heatwave risks (37).

Heatwave occurrence has increased across Africa over recent decades, although important regional heterogeneity exists in both exposure and vulnerability. Compared with many high-income settings, populations in sub-Saharan Africa face disproportionately higher risks owing to rapid urbanisation, widespread poverty, limited adaptive capacity and weaker health infrastructure (43). Evidence suggests that heatwaves are becoming more frequent, longer lasting and more intense across much of the continent, with urban areas experiencing particularly severe warming because of the urban heat island effect (18,44). Within low- and middle-income countries, heatwaves have been linked to increased demand for healthcare, reduced labour productivity, food insecurity and disruptions to essential services, highlighting the need for locally appropriate heat adaptation strategies (2,16,45,46). In Nigeria, recent analyses have documented increasing heatwave frequency and intensity, particularly across the northern regions, with projected future increases under continued climate warming (15,16). These events have been associated with impacts on agriculture, electricity demand, environmental degradation and public health, while rapid urbanisation and land-use change are expected to further amplify future heat exposure (16–18).

2.3 Heatwaves and public health

The health impacts of heatwaves are now well established. Numerous epidemiological studies have demonstrated associations between extreme heat and increased all-cause mortality, cardiovascular and respiratory disease, renal disease, adverse pregnancy outcomes, heat stroke and hospital admissions, with older adults, young children and individuals with chronic illnesses being particularly vulnerable (8–13). Beyond direct physiological effects, heatwaves may also disrupt healthcare delivery through increased pressure on health facilities, interruptions to electricity supply and cold-chain infrastructure, and reduced healthcare utilisation (6,7). These disruptions are particularly relevant for routine childhood immunization programmes, where reliable vaccine storage and timely attendance at health facilities are essential (47,48). Existing studies examining climate and vaccination have largely focused on vaccine storage, cold-chain integrity, seasonality and healthcare access, while relatively few have investigated whether exposure to extreme heat influences vaccination coverage or the timeliness of routine childhood immunization (46,49,50).

To date, evidence directly linking heatwave exposure to vaccine timeliness remains scarce, particularly in low- and middle-income countries and sub-Saharan Africa. This represents an important knowledge gap given the increasing frequency of heatwaves and the importance of timely vaccination in ensuring optimal protection against vaccine-preventable diseases.

3 Materials and Methods

3.1 Data

3.1.1 DHS Data

The 2024 Nigeria DHS survey (NDHS) was conducted between December 2023 and May 2024 by the National Population Commission (NPC) and was designed to provide nationally representative estimates for Nigeria as a whole, as well as for urban and rural areas, the six geopolitical zones, and each of the 36 states and the Federal Capital Territory (FCT). The 2024 NDHS employed a stratified two-stage cluster sampling design based on the updated cartographic sampling frame prepared for Nigeria's Population and Housing Census. The sampling frame excluded institutional populations such as persons in hotels, barracks, and prisons. In the first stage, enumeration areas (EAs), which served as primary sampling units or clusters, were selected with probability proportional to size within each of the 74 sampling strata defined by state and place of residence (urban/rural). A total of 1,400 clusters were selected, comprising 701 urban and 699 rural clusters. In the second stage, a household listing was conducted in each selected cluster, after which 30 households were selected using systematic sampling, yielding the intended sample of approximately 42,000 households(19).

Data for this study were obtained from the Women's Questionnaire of the 2024 NDHS. Women aged 15–49 years who were usual members of selected households or who had spent the night preceding the survey in the selected household were eligible for interview. The questionnaire collected information on respondents' sociodemographic characteristics, reproductive history, maternal and child health, family planning, and other health-related indicators. In total, 39,553 women were eligible for interview (19,166 from urban areas and 20,387 from rural areas), and a response rate of 99% was achieved nationally as well as in both urban and rural areas.

Information on childhood vaccination was obtained for children born in the three years preceding the survey, and alive at the time of the survey. For each eligible child, vaccination information, including receipt of each vaccine antigen and the corresponding date of administration, was extracted from the child's vaccination card where available. In the absence of a vaccination card, vaccination status was determined based on maternal recall. According to the 2024 NDHS, vaccination cards were seen for 50.2% of children aged 12–23 months and 34.4% of children aged 24–35 months, indicating that maternal recall constituted an important source of vaccination information for a substantial proportion of children(19).

3.1.2 Temperature data

Data on daily mean 2-metre air temperature for the period 1940–2024 were obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis via the Climate Data Store(51,52). The data are provided on a regular 0.25° × 0.25° latitude–longitude grid (approximately 27.8 kilometres squared). The data were extracted for a spatial bounding box enclosing Nigeria (west: 2.676932°E, south: 4.0690959°N, east: 14.678014°E, north: 13.885645°N) and retrieval was carried out programmatically in R using the `ecmwfr` package(53), which provides an interface to the ECMWF data services and the Climate Data Store API. The downloaded files were stored on secure local storage in NetCDF format, with one file saved for each year to facilitate efficient management and analysis.

3.2 Study population

The analysis was restricted to children born in the three years preceding the 2024 Nigeria Demographic and Health Survey. Children who had died before the survey were excluded since the DHS birth recode only collects vaccination information for living children and children who had missing information on date of birth were also excluded. For each vaccine antigen, the analysis included only children who had reached the recommended age of eligibility for that vaccine at the time of interview. For example, analyses of the first dose of the measles-containing vaccine (MCV1), which is scheduled at 9 months of age, were restricted to children aged at least 9 months. Therefore, the analytical sample differed slightly across vaccine antigens.

3.3 Outcome variables

The primary outcomes were delays in receipt of five key routinely recommended childhood vaccine antigens: Bacillus Calmette–Guérin (BCG), the first, second and third doses of the pentavalent vaccine (Penta1, Penta2 and Penta3), and the first dose of the measles-containing vaccine (MCV1). The pentavalent vaccine protects against the combination of diphtheria, tetanus, pertussis, hepatitis B and Haemophilus influenzae type b infections. According to Nigeria's Expanded Programme on Immunization (EPI) schedule(20), BCG is recommended at birth, Penta1 at 6 weeks, Penta2 at 10 weeks, Penta3 at 14 weeks, and MCV1 at 9 months. These antigens were selected because they represent key milestones in the infant immunization schedule and are widely used as indicators of the performance of routine immunization programmes during the first year of a child's life.

For each antigen, the recommended vaccination date was calculated by adding the scheduled age at vaccination to the child's date of birth. Vaccine delay was defined as receipt of a vaccine more than 28 days after its recommended due date. Vaccinations administered before or within 28 days of the scheduled date were classified as timely. Although no universally accepted definition of vaccination delay exists, previous studies have used thresholds ranging from one week to four weeks depending on the antigen and study setting(33). A threshold of 28 days was adopted in this study to provide a consistent definition across all vaccine antigens.

Vaccination dates recorded on vaccination cards were subjected to data quality checks based on the WHO recommended routine immunization schedule. Vaccination records were considered invalid if Penta1 was administered before 6 weeks of age, Penta2 before 10 weeks, Penta3 before 14 weeks, or MCV1 before 6 months of age. These checks were undertaken to identify the extent of implausible vaccination dates prior to analysis. Because vaccination dates were only available for children with documented vaccination cards, missing vaccination dates were common. Rather than excluding these children, missing vaccination times (defined as the number of days from birth to vaccination) were imputed using multiple imputation, allowing children without vaccination cards to contribute to the analysis. Details of the imputation procedure are provided in Section 2.8.

3.4 Exposure variable

There is in general no consensus as to the definition of a heatwave, and different countries, and studies adopt different definitions on what a heatwave is(11,37,54,55). A similarity across many definitions involves an aspect of duration, and intensity such as "a period of marked and unusually hot weather persisting for at least two consecutive days"(55) with common definitions being: a period of at least n (say 3 or 5) consecutive days where the temperature exceeds the local $x^{th}-$percentile (Say 90% or 95%). In this study, we adopt a definition and classificatin of heatwaves following the framework of probabilistic extreme event attribution(12,37,39), where we utilize a 5-day duration, and a 90%-tile threshold.

Daily mean 2-metre air temperature from ERA5 was used to derive heatwave exposure. The analysis was performed separately for each ERA5 grid cell covering Nigeria. To capture sustained periods of unusually high temperature and account for the accumulating effect of heat, a five-day rolling mean of daily temperature (Tx5x index) was calculated for every grid cell over the period January 1940 to May 2024. The resulting time series was divided into a baseline period (January 1940–December 2020) and an analysis period (January 2021–May 2024). The baseline period was used to estimate location-specific extreme temperature thresholds, while the analysis period was used to identify heatwave events during the period covered by the NDHS.

For each grid cell, the annual maximum of the Tx5x index was extracted from each year of the baseline period, yielding 81 annual maxima. These annual maxima were modelled using a non-stationary Generalized Extreme Value (GEV) distribution in which the location parameter varied linearly with calendar year to account for long-term warming trends(39):

$$M_{t,x} \sim GEV(\mu, \sigma, \zeta)$$

$$\mu = \beta_0 + \beta_1 \cdot t$$

where $M_{t,x}$ denotes the annual maximum Tx5x value at grid cell x in year t. The model was fitted using the `gamlss` and `gamlssx` packages in R(56,57).

The fitted GEV models were subsequently used to estimate Z_T, the 10-year return level for each grid cell. This return level represents the temperature expected to be exceeded, on average, once every ten years under the fitted non-stationary climate model. If $F(\cdot)$ denotes the fitted GEV cumulative distribution function(58):

$$F(x) = exp(-[1 + \zeta(\frac{x - \mu}{\sigma})]_+^{-1/\zeta})$$

Then, the T-year return level satisfies

$$P(M_{t,x} > Z_T) = \frac{1}{T}; F(Z_T) = 1 - \frac{1}{T}$$

For a 10-year return level, $F(Z_{10}) = 1 - 1/10 = 0.9$ and hence $Z_{10} = F^{-1}(0.9)$ which is given by:

$$Z_{10} = \mu + \frac{\sigma}{\zeta}[(-\log_e(0.9))^{-\zeta} - 1], \zeta \neq 0$$

Each Tx5x observation during the analysis period (January 2021–May 2024) was then compared with the corresponding grid-cell-specific return level. Days for which the Tx5x index exceeded the return level (Z_{10}) were classified as heatwave days, whereas all remaining days were classified as non-heatwave days. This produced a daily binary heatwave indicator for every ERA5 grid cell across Nigeria.

The DHS clusters were subsequently linked to the ERA5 heatwave data using their geographical coordinates and calendar date. Spatial linkage followed DHS recommendations for incorporating categorical spatial covariates by accounting for the dispacement distance. Because the ERA5 grid resolution (~0.25°, approximately 27 km) is substantially larger than the maximum DHS displacement distance (2km for urban clusters, 5km for 99% of rural clusters, 10km for 1% of rural clusters), each survey cluster was assigned the heatwave status of the ERA5 grid cell containing its displaced centroid. Finally, each observation was classified as heatwave-exposed if at least one heatwave day occurred within a 15-day exposure window centered on the scheduled vaccination date (7 days before to 7 days after).

3.5 Covariates

In this study, the covariates utilized were informed by a recent systematic review of factors associated with vaccine timeliness in low- and middle-income countries over 10 years (2007-2017)(33), and availability of the covariates in the DHS data. The covariates are shown in [xxx]:

3.6 Sensitivity analysis

We assessed the sensitivity of the final results to modelling choices namely: the heatwave classification model and the return level specified for the heatwave classification model. An alternative heatwave classification model, the Generalized Pareto Distribution was used to classify heatwaves using a methodology termed “Peaks-Over-Threshold” (POT). In this, instead of modelling annual maxima, such as in the GEV model, we model the extreme temperatures

Table 3.1: Description of predictor variables included in the analysis.

Variable	**Description**	**Type**
Gender	The child's gender	Binary
Delivery	The place of delivery categorized as either institutional or home delivery	Binary
Birth order	The child's birth order	Integer
Education	The highest level of education attained by the mother	Categorical
Age-at-birth	The age of the mother at birth	Continuous
ANC	The number of antenatal care visits attended by the mother	Continuous
DHF	The distance to the nearest health facility	Continuous
Residence	The residence, categorized as rural or urban	Binary
Wealth	The household wealth index categorized into 5 quintiles	Categorical

above a given threshold such as the 95% percentile, and use the model to classify whether future occuring events were heatwaves or not. Specifically, …

Further, we also assessed the sensitivity of the results, to the imputation scheme, by running a complete case analysis, to assess the extent to which the results are sensitive to the imputation.

gev stuff again, (return level: 15 year, 5 year) gpd stuff with 95%, 90% 99%

3.7 Statistical analysis

For any given antigen, we construct the appropriate dataset by only including children who have attained the antigen time of vaccination, and for a child i, we let Y_i represent the binary outcome variable for vaccine delay, which we code as 1 if the child has the vaccine delay, and 0 if not, Z_i represent the matrix of selected covariates, including the intercept, and X_i represent the binary exposure variable (heatwave), the model is given as follows:

$$Y_t \sim Bernoulli(\pi_i)$$

$$\text{log}(\frac{p_i}{1-p_i}) = Z_i \cdot \beta + X_i \cdot \gamma$$

We adjust for the survey's complex sampling scheme using the sampling weights in the model.

3.7.1 Statistical software

All analyses were conducted using R version 4.5.1(59). Data wrangling, manipulation and visualization were performed using the `tidyverse` suite of packages(60). Data retrieval and management were carried out using `ecmwfr`(61), `haven`(62), and `terra`(63), while spatial processing and analysis were performed using the `terra` and `sf` packages(63,64). Extreme value modelling was undertaken using `gamlss` and `gamlssx`(65,66), and Bayesian regression modelling, model diagnostics and posterior summarization were conducted using `brms`, `tidybayes`, and `bayesplot` interfaced with the Stan probabilistic programming language(67–69). Missing data were explored and handled using the `naniar` and `mice` packages(70,71). Reproducible reports and tables were generated using `rmarkdown` and `kableExtra`(72,73).

4 Results

5 Discussion

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Appendix